

Theorem 4.6.3. (Whitney 1932)

A graph is planar if and only if it has an abstract dual.

Proof. Let G be a planar graph, and consider any drawing. Every component C of this drawing has a plane dual C^* . Consider these C^* as abstract multigraphs, and let G^* be their disjoint union. Then the bonds of G^* are precisely those of the C^* , which by Proposition 4.6.1 correspond to the cycles in G .

(1.9.2)
(4.5.1)

Conversely, suppose that G has an abstract dual G^* . For a proof that G is planar, it suffices by Theorem 4.5.1 and Proposition 4.6.2 to show that $\mathcal{B}(G^*)$ has a sparse basis. By Proposition 1.9.2, it does. $\square$

The duality theory for both abstract and plane graphs can be extended to infinite graphs. As these can have infinite bonds, their duals must then have ‘infinite cycles’. Such things do indeed exist, and are fascinating: they arise as topological circles in a space formed by the graph and its *ends*; see Chapter 8.6.

Exercises

1. Show that every graph can be embedded in $\mathbb{R}^3$ with all edges straight.
2. ⁻ Show directly by Lemma 4.1.2 that $K_{3,3}$ is not planar.
3. Here is an inductive ‘proof’ that every maximal plane graph of order ≥ 4 is a plane triangulation of minimum degree 3. The induction starts with K^4 . For the induction step, consider an arbitrary maximal plane graph G of order $n \geq 4$, and consider all possible ways of extending it to a maximal plane graph G' of order $n+1$ by adding a new vertex v . No matter how this is done, v will come to sit in a face of G , which by the inductive assumption is bounded by a triangle. Since G' is maximally planar, v must be joined to all three vertices of that triangle. Clearly, G' is another plane triangulation, and $\delta(G') = d(v) = 3$.
 - (i) ⁻ Find the flaw in this ‘proof’.
 - (ii) Find a counterexample, and explain why the ‘proof’ overlooks it.
4. Show that every planar graph is a union of three forests.
5. The ancient Greeks loved regular plane graphs whose faces were bounded by cycles of the same length.
 - (i) Show that such graphs exist for only finitely many pairs (d, ℓ) of degree $d \geq 3$ and cycle length ℓ . Can you give an upper bound?
 - (ii) ⁺ Show that there are only finitely many such plane graphs, up to topological isomorphism.
6. A *fullerene* is a molecule that is made up entirely of carbon atoms forming a cubic plane graph all whose faces are pentagons or hexagons. Show that, since carbon atoms can form double bonds, every such graph can be realized in principle by (4-valent) carbon atoms.

7. A football is made of pentagons and hexagons, not necessarily of regular shape. They are sewn together so that their seams form a cubic planar graph. How many pentagons does the football have?
- 8.⁻ (continued from Exercises 6 and 7)
Fullerenes are less stable if they contain adjacent pentagons. Show that stable fullerenes have at least 60 carbon atoms.
9. Let G be a graph of order n that is embedded in a surface of Euler characteristic χ and cannot be embedded in a simpler surface (one of larger Euler characteristic). Show that G has at most $3n - 3\chi$ edges.
(Hint. You may use that every face of such an embedded graph is a topological disc. Such embeddings satisfy the general Euler formula, $n - m + \ell = \chi$.)
10. Find a direct proof for planar graphs of Tutte's theorem on the cycle space of 3-connected graphs (Theorem 3.2.6).
- 11.⁻ Show that the two plane graphs in [Figure 4.3.1](#) are not combinatorially (and hence not topologically) isomorphic.
12. Show that the two graphs in [Figure 4.3.2](#) are combinatorially but not topologically isomorphic.
- 13.⁻ Show that our definition of equivalence for planar embeddings does indeed define an equivalence relation.
14. Find a 2-connected planar graph whose drawings are all topologically isomorphic but whose planar embeddings are not all equivalent.
- 15.⁺ Show that every plane graph is combinatorially isomorphic to a plane graph whose edges are all straight.
(Hint. Given a plane triangulation, construct inductively a graph-theoretically isomorphic plane graph whose edges are straight. Which additional property of the inner faces could help with the induction?)

Do not use Kuratowski's theorem in the following two exercises.

16. Show that any minor of a planar graph is planar. Deduce that a graph is planar if and only if it is the minor of a grid. (*Grids* are defined in Chapter 12.4.)
17. (i) Show that the planar graphs can in principle be characterized as in Kuratowski's theorem: that there exists a set $\mathcal{X}$ of graphs such that a graph G is planar if and only if G has no minor in $\mathcal{X}$.
(ii) Can every graph property be characterized in this way? If not, which can?
18. Does every planar graph have a drawing with all inner faces convex?
19. Modify the proof of Lemma 4.4.3 so that all inner faces become convex.
20. Does every minimal non-planar graph G (i.e., every non-planar graph G whose proper subgraphs are all planar) contain an edge e such that $G - e$ is maximally planar? Does the answer change if we define 'minimal' with respect to minors rather than subgraphs?

21. Show that adding a new edge to a maximal planar graph of order at least 6 always produces both a TK^5 and a $TK_{3,3}$ subgraph.
22. Prove the general Kuratowski theorem from its 3-connected case by manipulating plane graphs, i.e. avoiding Lemma 4.4.5.
(This is not intended as an exercise in elementary topology; for the topological parts of the proof, a rough sketch will do.)
- 23.⁻ A graph is called *outerplanar* if it has a drawing in which every vertex lies on the boundary of the outer face. Show that a graph is outerplanar if and only if it contains neither K^4 nor $K_{2,3}$ as a minor.
24. Show that a 2-connected plane graph is bipartite if and only if every face is bounded by an even cycle.
25. Let $G = G_1 \cup G_2$, where $|G_1 \cap G_2| \leq 1$. Show that $\mathcal{C}(G)$ has a sparse basis if both $\mathcal{C}(G_1)$ and $\mathcal{C}(G_2)$ have one.
26. Find a cycle space basis among the face boundaries of a 2-connected plane graph.
27. Show that a 3-connected graph of order n has at least $n/2$ peripheral cycles. Is this lower bound sharp?
- 28.⁺ Find an algebraic proof of Euler's formula for 2-connected plane graphs, along the following lines. Define the *face space* $\mathcal{F}$ (over $\mathbb{F}_2$) of such a graph in analogy to its vertex space $\mathcal{V}$ and edge space $\mathcal{E}$. Define *boundary maps* $\mathcal{F} \rightarrow \mathcal{E} \rightarrow \mathcal{V}$ in the obvious way, specifying them first on single faces or edges (i.e., on the standard bases of $\mathcal{F}$ and $\mathcal{E}$) and then extending these maps linearly to all of $\mathcal{F}$ and $\mathcal{E}$. Determine the kernels and images of these homomorphisms, and derive Euler's formula from the dimensions of those subspaces of $\mathcal{F}$, $\mathcal{E}$ and $\mathcal{V}$.

A family of subgraphs of G is said to form a *double cover* of G if every edge of G lies in exactly two of those subgraphs. A double cover by cycles is a *cycle double cover*. A *surface map* is a graph embedded in a surface so that every face is bounded by a cycle.

29. (for topologists)
 - (i) Show that a cubic graph has a cycle double cover if and only if it is isomorphic to a surface map.
 - (ii) Does every cycle double cover of a cubic graph occur as the family of face boundaries of a surface graph isomorphic to it?
 - (iii) Is the assumption of being cubic relevant in (ii)?

Given a 2-connected graph $G = (V, E)$ and an integer χ , call another graph $H = (W, E)$ with the same edge set E as G an *abstract dual of G with parameter χ* if all the atomic bonds of G are cycles in H and $|V| - |E| + |W| = \chi$.

- 30.⁺ (continuing Exercise 29)
Show that a graph is isomorphic to a surface map for a surface of Euler characteristic χ if and only if it has an abstract dual with parameter χ . You may use the results from Appendix B as needed.

31. Let G be a 2-connected graph whose cycle space is generated by a sparse set $\mathcal{C}'$ of cycles. From MacLane's theorem we know that G even admits a double cover by cycles generating $\mathcal{C}(G)$: the face boundaries in any drawing of G . Show directly (without using MacLane's theorem) that $\mathcal{C}'$ extends to a cycle double cover $\mathcal{D}$ of G .
- 32.⁺ (for topologists) Prove the non-trivial implication in MacLane's theorem constructively, as follows. Assume that the given graph G is 2-connected and, by the previous exercise, has a double cover $\mathcal{D}$ by cycles generating $\mathcal{C}(G)$. For each of these cycles C take a disc and identify its boundary with C .
 - (i) Show that the space obtained is a surface, i.e., a compact 2-manifold without boundary.
 - (ii) Use Theorem 1.9.5 to show that this surface has Euler characteristic at least 2. (This implies that it must be the sphere, a fact you may assume as known.)
33. Deduce from the last two exercises that, given any 2-connected planar graph and a sparse basis $\mathcal{C}'$ of $\mathcal{C}(G)$ consisting of cycles, there is a drawing of G in which the cycles in $\mathcal{C}'$ are precisely the boundaries of the inner faces.
- 34.⁺ Let C be a closed curve in the plane that intersects itself at most once in any given point of the plane, and where every such self-intersection is a proper crossing. Call C *alternating* if we can turn these crossings into over- and underpasses in such a way that when we run along the curve the overpasses alternate with the underpasses.
 - (i) Prove that every such curve is alternating, or find a counterexample.
 - (ii) Does the solution to (i) change if the curves considered are not closed?
- 35.⁻ What does the plane dual of a plane tree look like?
- 36.⁻ Show that the plane dual of a plane multigraph is connected.
- 37.⁺ Show that a connected plane multigraph has a plane dual.
38. Show that any two plane duals of a plane multigraph are combinatorially isomorphic.
39. Let G^* be an abstract dual of G , and let $e = e^*$ be an edge. Prove the following two assertions:
 - (i) G^*/e^* is an abstract dual of $G - e$.
 - (ii)⁺ $G^* - e^*$ is an abstract dual of G/e .
40. Find a connected graph that has two non-isomorphic abstract duals. Can you find a 2-connected example?
41. Let G, G^* be dual plane graphs. Prove the following statements:
 - (i) If G is 2-connected, then G^* is 2-connected.
 - (ii) If G is 3-connected, then G^* is 3-connected.
 - (iii) If G is 4-connected, then G^* need not be 4-connected.

42. Give detailed proofs for the statements made after Proposition 4.6.2, except for the uniqueness of G^* (which is proved in Exercise 43 (ii)).
43. Let $G^* = (V^*, E^*)$ be a connected abstract dual of a connected multigraph $G = (V, E)$. Does G have a drawing whose plane dual is isomorphic to G^* ? (Perhaps even ‘canonically’ isomorphic? In which sense?)
 - (i) For 2-connected G , prove this using the approach of Exercise 32.
 - (ii) Deduce that abstract duals of 3-connected graphs are unique. (What exactly could this mean? Suggest a definition of uniqueness that is stronger than ‘up to isomorphism’.)
 - (iii) Find a counterexample to the general statement.
44. Show that the following statements are equivalent for connected multigraphs $G = (V, E)$ and $G' = (V', E)$ with the same edge set:
 - (i) G and G' are abstract duals of each other;
 - (ii) given any set $F \subseteq E$, the multigraph (V, F) is a tree if and only if $(V', E \setminus F)$ is a tree.

Notes

There is a very thorough monograph on the embedding of graphs in surfaces, including the plane: B. Mohar & C. Thomassen, *Graphs on Surfaces*, Johns Hopkins University Press 2001. Proofs of the results cited in Section 4.1, as well as all references for this chapter, can be found there. A good account of the Jordan curve theorem, both polygonal and general, is given also in J. Stillwell, *Classical topology and combinatorial group theory*, Springer 1980.

The short proof of Corollary 4.2.10 uses a trick that deserves special mention: the so-called *double counting* of pairs, illustrated in the text by a bipartite graph whose edges can be counted alternatively by summing its degrees on the left or on the right. Double counting is a technique widely used in combinatorics, and there will be more examples later in the book.

The material of Section 4.3 is not normally standard for an introductory graph theory course, and the rest of the chapter can be read independently of this section. However, the results of Section 4.3 are by no means unimportant. In a way, they have fallen victim to their own success: the shift from a topological to a combinatorial setting for planarity problems which they achieve has made the topological techniques developed there dispensable for most of planarity theory.

In its original version, Kuratowski’s theorem was stated only for topological minors; the version for general minors was added by Wagner in 1937. Our proof of the 3-connected case (Lemma 4.4.3) is a weakening of a proof due to C. Thomassen, Planarity and duality of finite and infinite graphs, *J. Comb. Theory, Ser. B* **29** (1980), 244–271, which yields a drawing in which all the inner faces are convex (Exercise 19). The existence of such ‘convex’ drawings for 3-connected planar graphs follows already from the theorem of Steinitz (1922) that these graphs are precisely the 1-skeletons of 3-dimensional convex polyhedra. Compare also W.T. Tutte, How to draw a graph, *Proc. Lond. Math. Soc.* **13** (1963), 743–767.